

Sze Yoong Low

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EDUCATION

Imperial College London

London, United Kingdom

Bachelor of Engineering in Computing

Sep. 2025 – June 2028

- Dean's List (Year 1): Top 10% in cohort of 220
- First-Class Honours (Year 1): Achieved 88.47% with high First-Class grades in all eight modules, including Computing Practical 1 (93%), Graphs and Algorithms (92%), Discrete Mathematics, Logic, and Reasoning (92%)

EXPERIENCE

Maxis (*Bursa Malaysia 6012*)

Kuala Lumpur, Malaysia

AI Automation Software Engineering Intern

July 2026 – Sep. 2026

- Revamped an in-house agent orchestrator and observability dashboard to collect token use telemetry, give business users self-service real-time analytics reports, and automate AWS Step Function construction and deployment
- Managed production deployments of serverless container apps, AWS Lambda functions, and database migrations
- Redesigned the database, REST API, and admin console of an internal LLM reverse proxy to improve cost accounting accuracy by capturing nuances like KV caching and enable request routing to the lowest cost provider

Boston Consulting Group (BCG)

Kuala Lumpur, Malaysia

Case Team Assistant

March 2025 – June 2025

- Provided client with clarity on group-wide digital spending by building an Excel dashboard to consolidate siloed data sources and visualise key metrics tailored to both operational staff and senior management

PROJECTS

Canary | *Agentic Jupyter Notebook for building interactive analyst reports*

June 2026 – Present

- Assembled a bespoke Terraform DevOps workflow to automate the lifecycle of isolated AWS and Cloudflare development environments provisioned on-demand, and promote artifacts to production without rebuilding
- Developed a modular Polars data pipeline with full flexibility to perform any logical composition of columnar data transformations while deduplicating lazy data fetches by planning executions as directed acyclic dependency graphs
- Created an AI agent with LangGraph to reliably translate users' questions into financial terminal function calls
- Enabled users to save and share reports with robust access controls enforced by server-side access token authorisation checks against privileges that admins and content owners can adjust instantly with full audit history
- Minimised network bandwidth required for client-server syncs by maintaining client state in TanStack query caches

3D Cellular Automata Simulation Engine | *GPU acceleration, voxel optimisations*

May 2026 – June 2026

- Led my team of four to develop a C simulator for real-world systems such as organism growth and galaxy filaments
- Built an OpenGL graphics pipeline that renders a million voxels at over 100 FPS by using compute shaders to perform stream reduction optimisations including frustum and occlusion culling natively in GPU memory
- Boosted the frame rate of simulations with transparent effects four to ten fold by refining culling algorithms to target individual cube faces and replacing the log-linear bitonic merge sorting used to prevent Z-fighting with a constant-time Weighted Blended Order-Independent Transparency algorithm
- Presented our work to engineers from Arm and fielded audience questions as finalists for Best First-Year Project

RoboCup Small Size League | *Real-time digital signal processing for autonomous robots*

Oct. 2025 – June 2026

- Software engineer with Imperial's First Order Robotics team that competed for a spot at the World Finals
- Improved motion planning accuracy by 35% by implementing multivariate Kalman filters with NumPy to minimise noise in position data feeds from overhead cameras and predict missing values
- Enabled intuitive and convenient simulation result benchmarking and parameter tuning by building a dashboard of maps and charts in Jupyter Notebook with Matplotlib and Pandas
- Equipped match simulator to stress test software under real-world conditions like random noise and missing data

SKILLS AND TECHNOLOGIES

Programming: C, Kotlin, Java, Python, TypeScript, Haskell

Infrastructure and DevOps: AWS, Terraform, Docker, Cloudflare Workers, GitHub Actions, Bash, PowerShell

Full-stack web development: FastAPI, Express.js, Flask, LangGraph, Auth0, React, React Router, TanStack

Databases and analytics: PostgreSQL, MySQL, SQLite, Polars, Pandas, NumPy, Matplotlib, Apache ECharts, Excel

Native languages: Mandarin Chinese, Cantonese, Bahasa Melayu